

Grammatical Inference Applications in Bioinformatics

Progress Report

Malcolm Agius
malcolm.agius.19@um.edu.mt
University of Malta

Abstract

Deterministic Finite-State Automaton (DFA) Learning is an area of Machine Learning, specifically Grammatical Inference, that involves inferring a regular language by finding the minimum-state DFA that is consistent with a set of positive and negative example strings. This research focuses on the problem of classifying amyloidogenic hexapeptides. Amyloids are proteins that are capable of forming fibrils, rather than functioning in their intended manner, and are associated with a group of diseases called amyloidosis. Amyloidosis diseases include Alzheimers disease, Huntingtons disease, Parkinson's disease and type II diabetes.

This project will be focusing on recreating the results achieved by Wieczorek and Unold in 2014 and 2016 on the state-of-the-art DFA learning algorithms using EDSM as a baseline, and applying more recent algorithms in this problem, attempting to improve upon EDSM and the other algorithms tested by Wieczorek and Unold.

1 Introduction

Grammatical Inference is a field of study where a formal grammar is inferred from a set of examples of strings that belong to a language and, generally, another set of examples of strings that do not belong to language. An area of Grammatical Inference is Deterministic Finite-State Automaton (DFA) learning where a minimum-state DFA is induced from a set of positive and negative sample strings.

In this project, we will be applying DFA learning techniques to a bioinformatics problem, specifically, the classification of amyloidogenic hexapeptides. Hexapeptides are protein residues consisting of a sequence of six amino acids.

The language that represents all possible hexapeptides is a finite language, since the length of each string and the size of the alphabet are finite numbers. Therefore, the language of amyloidogenic hexapeptides, being a subset of the language of hexapeptides, is also finite. This implies that both languages are regular. DFA learning is a suitable way to approach this problem since the target language is a regular language. Thus, considering each hexapeptide as a string and trying to find a DFA that correctly classifies unseen hexapeptides by learning from a data set of hexapeptides labelled amyloid/non-amyloid reduces this problem into a DFA learning problem.

1.1 Motivation

Amyloids are proteins that may of form fibrils rather than functioning in the way that they are intended. They are associated with a group of diseases called amyloidosis, which includes diseases such as Alzheimers, Huntingtons disease, and type II diabetes [14]. It is also believed that short regions of proteins such as hexapeptides are solely able to form amyloid-like fibrils [11].

Therefore, the incentive to detect amyloidogenic hexapeptides computationally is to find which peptides in a patient's anatomy the disease is originating from, without needing to experimentally classify the hexapeptides.

1.2 Aims & Objectives

The aim of this project is to help with the bioinformatics problem of classifying amyloidogenic hexapeptides by researching and experimentally testing DFA learning algorithms.

The objectives for this project are as follows:

- Research current state-of-the-art DFA learning algorithms and understand how they work.
- Familiarise ourselves with the field of bioinformatics and understand the problem we are tackling.
- Replicate a selection of algorithms, such as EDSM, that were implemented by Wieczorek and Unold in 2016 [15] on the same data, as a baseline.
- Implement more novel algorithms and compare their performance to the aforementioned baseline.

1.3 Why the Problem is Non-Trivial

Testing amyloidogenicity of all possible combinations of hexapeptides experimentally is currently not feasible [14]. Therefore, several computational methods have been developed to predict the class [14].

In this case, the problem has been reduced to a DFA learning problem. The problem of finding the minimum-state DFA that is consistent with a finite set of positive and negative sample strings is NP-Complete [4].

2 Background

A DFA is formally defined as a 5-tuple $(Q, \Sigma, \delta, q_0, F)$, where Q is the set of states, Σ is the alphabet, δ is the transition function, $q_0 \in Q$ is the start state and $F \subseteq Q$ is the set of final states. A finite-state automaton is deterministic when for all the states in the automaton, from each state there exists at most one transition for every label. In this research we

will tackle this problem by learning deterministic automata, however, in the past, studies have been done tackling this problem by learning non-deterministic finite-state automata (NFAs) [14].

The DFA Learning Problem is the problem of finding the minimum-state DFA consistent with a training set S of positive examples (S_+) and negative examples (S_-), where $S = S_+ \cup S_-$.

The algorithms that exist in DFA Learning can be categorised into two classes, exact algorithms that find the exact minimum-state DFA, and heuristic methods that try to find an accurate estimate of what that target DFA should be [14]. Given that the problem is NP-Complete, exact methods are not feasible to be used and in this research we will focus on the heuristic based methods.

Regarding the technique of using heuristic methods, many DFA Learning algorithms are said to be state merging algorithms, one of which is Rodney Price's Abbadingo winning heuristic of Evidence-Driven State Merging (EDSM) [8]. State merging algorithms usually start by building an Augmented Prefix Tree Acceptor (APTA, Fig. 1), which is a DFA that accepts/rejects exactly the strings in the training set, and cannot give any classification for unseen strings. Once the APTA is built, state merging algorithms attempt to find a partition of the set Q of states of the APTA, where each block in the partition represents a state in the hypothesised DFA of the algorithm, which is a more general DFA that can generalise for unseen data. In these algorithms, the partition of set Q is usually built by merging different states together, which turns the set Q of states in the APTA into a partition where the states that have been merged together make up a block in the new partition. Therefore, different state merging algorithms simply differ in the heuristic they use that chooses which states should be merged together.

It is good to note that state merging algorithms are not the only ways of tackling this problem. The problem has also been reduced to the graph colouring problem [2] and the Satisfiability problem [7], amongst other approaches.

The field of Bioinformatics is a subdiscipline of computer science, biology, mathematics and statistics, most often dealing with the analysis of DNA and amino acid sequences. Hexapeptides are chains of six amino acids. A hexapeptide can be represented as sequence of six characters, each character representing a different amino acid. The protein alphabet contains 20 characters, therefore in our problem $\Sigma = \{A, C, D, E, F, G, H, I, K, L, M, N, P, Q, R, S, T, V, W, Y\}$ and $|\Sigma| = 20$.

These properties make the target language in our problem a finite language, making it regular. This means that the target language can be represented by a DFA, which is the representation we will use for our hypothesised language.

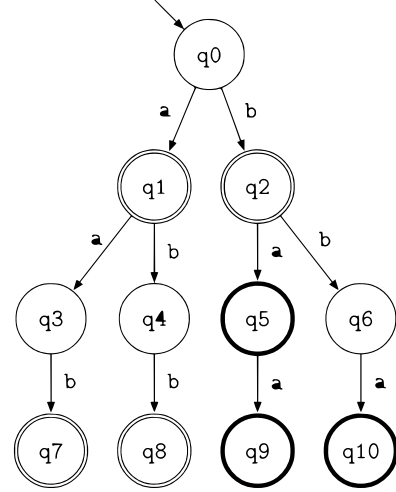


Figure 1. This is the APTA for a training set S having $S_+ = \{a, b, aab, abb\}$ and $S_- = \{ba, bba, baa\}$ [5]

3 Literature Review

The first theoretical contribution to the area of Grammatical Inference was done by Gold in 1967 [3]. He identified the very first model, identification in the limit, which works by iteratively taking a string from an oracle or from a training set and each time updating the hypothesised language to satisfy the new string. Due to this method, Gold identifies a class of languages that he calls the class of languages identifiable in the limit. This class of languages is the class of languages that can be exactly identified from a set of finite examples in the limit.

In 1978 [4], Gold also showed that the DFA Learning problem of finding a minimum-state DFA from a finite set of positive and negative strings is NP-Complete. Due to the problem being NP-Complete, many studies moved away from trying to identify the minimum-state DFA exactly and resorted to a new learning model, which was introduced by Valiant in 1978, PAC Learning [12]. PAC (Probably Approximately Correct) Learning is a learning model where it is allowed that the target hypothesis is not exactly the target concept. Due to the NP-Completeness of the DFA Learning problem, and due to the limited data that is available on amyloidogenic hexapeptides, we will use algorithms that use the PAC Learning Model to tackle this problem.

Oncina and Garcia, in 1992, develop an algorithm called RPNI (Regular Positive and Negative Inference) [10], which uses positive and negative example strings in order to attempt to find the minimum-state DFA. This algorithm is in the class of state merging algorithms, because it learns a DFA by merging states iteratively, starting from building an APTA, while making sure that the resultant DFA is still consistent with the training set, until no more merges are possible (i.e. any other merge would make the DFA inconsistent with training data).

In 1997, the Abbadingo One Competition [8] was held. This competition was held by Lang and Permuter, and its purpose was to encourage development of new algorithms that tackle the DFA learning problem, with an emphasis on having sparse data sets. The competition turned out to be fruitful and was won by Rodney Price’s Evidence Driven State Merging (EDSM) algorithm, which to this day is considered as a baseline DFA learning algorithm for comparison when conducting new research [18]. The competition was concerned with learning DFAs of nominal size of 64, 128, 256 or 512 states. In order for a problem to be solved, the hypothesised DFA had to correctly identify more than 99% of strings from a test set of 1800 strings.

A more recent competition, the StaMinA Competition held by Walkinshaw et al. [13], took place, which had a very similar setup to the Abbadingo One Competition. The StaMinA Competition focuses on the complexity of learning with respect to the alphabet size, and tests the algorithms on alphabet sizes of 2, 5, 10, 20 and 50. The winning algorithm was the DFASAT algorithm, by Marijn Heule and Sicco Verwer. Heule and Verwer’s idea was to Turing Reduce the DFA Learning problem into the Satisfiability (SAT) Problem, and then take advantage of the much more developed and mature algorithms to solve the problem using SAT Solvers. It is interesting to note that in this competition, problems instances where the alphabet size was 20 were given, and DFASAT performed the best in these. This is interesting because in fact, the problem of classifying amino acid sequences, co-incidentally also has an alphabet size of 20, as mentioned in Section 2. However, it does not seem that this method of DFA learning has been applied and tested on our specific problem.

In recent times, empiricists have developed certain Grammatical inference algorithms to specifically solve bioinformatics tasks, rather than just the theoretical DFA learning problem. In fact, in 2006, Coste and Kerbellec [1] implemented an algorithm that learnt automata for protein families, but not amyloids. The authors stated that given a labelled training set of protein sequences in a certain family (positive data only), their algorithm can infer an automaton that approximately represents the protein family. The method they built is inspired by both grammatical inference techniques and protein multiple alignment techniques. This approach takes a characterisation stage that uses protein alignment techniques to identify the most similar and representative parts between the protein set, and generalisation stage which builds an automaton by merging the representative parts.

Wieczorek and Unold tackle the precise problem of classifying amyloidogenic hexapeptides using Grammatical Inference techniques in 2014 [14]. However, in this case, their algorithm learns an NFA as apposed to a DFA, specifically a Directed Acyclic Word Graph (DAWG). Their algorithm consists of two phases. The first phase involves building an

automaton in a similar way to that of a minimal DFA, but allowing for non-determinism. In the second phase, the graph built in the first phase is generalised by extending it in order to accept more words. In their evaluation, the algorithm was compared with other established Grammatical Inference Algorithms, one of which was EDSM. It was shown that the DAWG algorithm achieves better balanced accuracy and recall than the other algorithms but lower precision.

Two years later, in 2016 [15], Wieczorek and Unold propose a new approach for using Grammatical Inference to solve this problem, attempting to improve upon their previous DAWG implementation. Their algorithm focuses on building star-free regular expressions (SFRE). In essence, an SFRE is a regular expression that does not contain Kleene Closures, hence being star-free. In contrast to the DAWG implementation, the SFRE implementation got less recall than its counterparts, but achieved the best specificity (true negative rate). The precision metric was not used in this study.

Wieczorek and Unold, together with Strak, developed a new method in 2019 that generates an acyclic word graph [17]. They propose the idea of a Suffix Classification Tree (SCT) and an algorithm that generates one from training data containing positive and negative examples. A brief description of an SCT is a rooted tree where each of its edges is labelled by a substring, and each vertex is labelled by two factor frequencies, which are the frequency of a factor being present in positive examples, and the frequency of the factor being present in negative examples. Concatenating every string in the path from the root to a given node gives a substring of a string s where s is a positive or negative example. When given an unseen string x , the output of this method gives a real number between 0 and 1, which is a ratio between the number of substrings of a given length k in x that are also substrings of positive strings and the total number of strings in the training set that contain substrings of x . In this paper, the authors use the Area Under the Receiver Operating Characteristic Curve (AUC) metric, which shows how well the two classes, amyloid and non-amyloid have been separated. It can be seen that the SCT method perform much better than the other algorithms (such as EDSM) according to the AUC metric (AUC of 0.805 from SCT and 0.576 from Blue-fringe EDSM).

In the same year, Wieczorek and Unold [16] also published a paper that solves the same problem using Grammatical Inference. However, in this case, they propose a Genetic Programming technique that infers a noncircular context-free grammar (CFG), rather than a regular grammar, which is the class of grammars that can be expressed by a DFA/NFA. It is shown that from the combined metrics used in this study, this algorithm is on top of all other implementations for F1 score and AUC, and only beaten by one other implementation in Matthews Correlation Coefficient (MCC).

Most recently, in 2020, Wieczorek et al. [18] developed a new grammar inference method that infers non-circular parsing expression grammars (PEG), and applied it to our problem. The authors claim that PEGs are as fast as Finite-State Automata but can express more than CFGs. This algorithm is shown to have the best accuracy, AUC and MCC from all the compared algorithms in the paper.

4 Proposed Solution

In this research, the bioinformatics task of classifying amyloidogenic hexapeptides will be tackled using DFA learning techniques. Our plan is to use the online available WALTZ-DB data set [9] of 1416 labelled hexapeptides (515 positive and 901 negative) that was also used in several other studies [15] [18].

We plan on first applying a select few state-of-the-art algorithms such as different variants of EDSM and compute metrics that analyse their performance on the data set. Then we use the metrics gained on EDSM as a baseline gold standard on which we could compare other algorithms. We will attempt to recreate an approximation of the performance metrics used in other research in order to make sure that the approach that we are taking is a fair approach compared to other studies.

The next step would be to implement novel algorithms in DFA learning that are yet to be applied to this problem and compare the performance metrics with the baseline we would have created previously. Having this comparison would then allow us to also compare the algorithms newly introduced to this problem with algorithms that were discussed in other literature. We may also modify existing algorithms or create new algorithms specifically for this problems and also compare the metrics achieved to the baseline.

All the code used in this project will be written in the Go programming language. The DfaGo package [6] will be utilised to ease the process of conducting the required experiments for this project.

5 Evaluation Plan

As mentioned in Section 4, we will first implement state of the art DFA Learning algorithms such as EDSM and calculate certain metrics for its performance on the classification of a test set. Then we will evaluate other algorithms by computing the same metrics and comparing them directly to the baseline algorithm.

The metrics that can be used to evaluate the performance of our algorithms are:

- Accuracy

$$Accuracy = \frac{tp+tn}{(tp+tn+fp+fn)}$$

Accuracy is used widely in research to evaluate the performance of algorithms, such as in the Abbadingo

One Competition, and is generally indicative of the overall quality of a model.

- Precision

$$Precision = \frac{tp}{(tp+fp)}$$

Precision will give us the level of confidence one can have in the model inferred, when it classifies a hexapeptide as amyloidogenic.

- Negative predictive value (NPV)

$$NPV = \frac{tn}{(tn+fn)}$$

Similarly to precision, NPV will give us the level of confidence one can have when the model classifies a hexapeptide as non-amyloid.

- F1 Score

$$F1 = \frac{2 \times (Precision \times Recall)}{(Precision + Recall)}$$

F1 Score is a good metric to use in this case since we are dealing with a data set with 515 positive examples and 901 negative examples. F1 score is generally a good metric to use when the classes are imbalanced.

6 Conclusion

In this report we discuss the progress that has been done so far for this project. We define the problem we want to solve, i.e. classification of amyloidogenic hexapeptides, and we define our aims of finding new algorithms that improve upon previous work on this problem, with a focus on work by Wieczorek and Unold in 2014 and 2016 [14] [15]. We then give an overview of the fundamental background knowledge needed to follow through this report. We also give a description of the literature that inspired us to tackle this problem, and literature that helped us solidify our background knowledge in DFA Learning, that will be of utmost importance to achieve our aims for this project. A brief description of our methodologies for the proposed solution followed, along with an evaluation plan that will assess the performance of the algorithms we will implement.

References

- [1] François Coste and Goulven Kerbellec. 2006. Learning Automata on Protein Sequences. In *JOBIM*, Alain Denise, Pascal Durrens, Stéphane Robin, Eduardo Rocha, Antoine de Daruvar, and Alexis Groppi (Eds.). Bordeaux, France, 199–210. <https://hal.inria.fr/inria-00180429>
- [2] François Coste and Jacques Nicolas. 1997. Regular Inference as a graph coloring problem. In *In Workshop on Grammatical Inference, Automata Induction, and Language Acquisition (ICML'97)*.
- [3] E Mark Gold. 1967. Language identification in the limit. *Information and Control* 10, 5 (1967), 447–474. [https://doi.org/10.1016/S0019-9958\(67\)91165-5](https://doi.org/10.1016/S0019-9958(67)91165-5)
- [4] E Mark Gold. 1978. Complexity of automaton identification from given data. *Information and Control* 37, 3 (1978), 302–320. [https://doi.org/10.1016/S0019-9958\(78\)90562-4](https://doi.org/10.1016/S0019-9958(78)90562-4)
- [5] Kristian Guillaumier. 2020. *Learning DFAs from Sparse Data*. dissertation. University of Malta.

- [6] Kristian Guillaumier. 2021. *Practical DFA Learning with DfaGo*. University of Malta, Msida, Malta.
- [7] Marijn J. H. Heule and Sicco Verwer. 2010. Exact DFA Identification Using SAT Solvers. In *Proceedings of the 10th International Colloquium Conference on Grammatical Inference: Theoretical Results and Applications* (Valencia, Spain) (*ICGI'10*). Springer-Verlag, Berlin, Heidelberg, 66–79.
- [8] Kevin J. Lang, Barak A. Pearlmutter, and Rodney A. Price. 1998. Results of the Abbadingo one DFA learning competition and a new evidence-driven state merging algorithm. In *Grammatical Inference*, Vasant Honavar and Giora Slutzki (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 1–12.
- [9] Nikolaos Louros, Katerina Konstantoulea, Matthias De Vleeschouwer, Meine Ramakers, Joost Schymkowitz, and Frederic Rousseau. 2019. WALTZ-DB 2.0: an updated database containing structural information of experimentally determined amyloid-forming peptides. *Nucleic Acids Research* 48, D1 (09 2019), D389–D393. <https://doi.org/10.1093/nar/gkz758> arXiv:<https://academic.oup.com/nar/article-pdf/48/D1/D389/31697694/gkz758.pdf>
- [10] J. Oncina and P. Garcia. 1992. Inferring regular languages in polynomial update time. In *Pattern Recognition and Image Analysis*, N. Pérez de la Blanca, A. Sanfeliu, and E. Vidal (Eds.). Series in Machine Perception and Artificial Intelligence, Vol. 1. World Scientific, Singapore, 49–61.
- [11] Michael Thompson, Stuart Sievers, John Karanicolas, Magdalena Ivanova, David Baker, and David Eisenberg. 2006. The 3D profile method for identifying fibril-forming segments of proteins. *Proceedings of the National Academy of Sciences of the United States of America* 103 (04 2006), 4074–8. <https://doi.org/10.1073/pnas.0511295103>
- [12] L. G. Valiant. 1984. A Theory of the Learnable. *Commun. ACM* 27, 11 (nov 1984), 1134–1142. <https://doi.org/10.1145/1968.1972>
- [13] Neil Walkinshaw, Bernard Lambeau, Christophe Damas, Kirill Bogdanov, and Pierre Dupont. 2013. STAMINA: A competition to encourage the development and assessment of software model inference techniques. *Empirical Software Engineering* 18 (08 2013). <https://doi.org/10.1007/s10664-012-9210-3>
- [14] Wojciech Wieczorek and Olgierd Unold. 2014. Induction of Directed Acyclic Word Graph in a Bioinformatics Task. In *The 12th International Conference on Grammatical Inference (Proceedings of Machine Learning Research, Vol. 34)*, Alexander Clark, Makoto Kanazawa, and Ryo Yoshinaka (Eds.). PMLR, Kyoto, Japan, 207–217. <https://proceedings.mlr.press/v34/wieczorek14a.html>
- [15] Wojciech Wieczorek and Olgierd Unold. 2016. Use of a Novel Grammatical Inference Approach in Classification of Amyloidogenic Hexapeptides. *Computational and Mathematical Methods in Medicine* 2016 (2016). <https://doi.org/10.1155/2016/1782732>
- [16] Wojciech Wieczorek and Olgierd Unold. 2019. GP-Based Grammatical Inference for Classification of Amyloidogenic Sequences. In *Computational Intelligence Methods for Bioinformatics and Biostatistics*, Massimo Bartoletti, Annalisa Barla, Andrea Bracciali, Gunnar W. Klau, Leif Peterson, Alberto Policriti, and Roberto Tagliaferri (Eds.). Springer International Publishing, Cham, 81–91.
- [17] Wojciech Wieczorek, Olgierd Unold, and Łukasz Strak. 2019. Suffix Classification Trees. In *Proceedings of The 14th International Conference on Grammatical Inference 2018 (Proceedings of Machine Learning Research, Vol. 93)*, Olgierd Unold, Witold Dyrka, and Wojciech Wieczorek (Eds.). PMLR, 44–53. <https://proceedings.mlr.press/v93/wieczorek19a.html>
- [18] Wojciech Wieczorek, Olgierd Unold, and Łukasz Strak. 2020. Parsing Expression Grammars and Their Induction Algorithm. *Applied Sciences* 10, 23 (2020). <https://doi.org/10.3390/app10238747>